



OPEN ACCESS

APPROVED BY
Frontiers Editorial Office,
Frontiers Media SA, Switzerland

*CORRESPONDENCE
Collins Nduka Esomchi,
✉ ndukakuee@gmail.com

RECEIVED 04 September 2026
ACCEPTED 07 September 2026
PUBLISHED 17 September 2026

CITATION

Okposhi FT, Esomchi CN, Onwe KT, Onumajuru LO, Omeje EM, Onwumere FC, Ugwu PC, Onukogu GE, Olisaeke JO, Victor CP, Omeh CP and Onele BA (2026) Correction: Chronic exposure to mining-contaminated borehole water impairs the entire male reproductive axis: partial mitigation by alpha-lipoic acid in Wistar rats. *Front. Toxicol.* 8:1988528. doi: 10.3389/ftox.2026.1988528

COPYRIGHT

© 2026 Okposhi, Esomchi, Onwe, Onumajuru, Omeje, Onwumere, Ugwu, Onukogu, Olisaeke, Victor, Omeh and Onele. This is an open-access article distributed under the terms of the [Creative Commons Attribution License \(CC BY\)](#). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Correction: Chronic exposure to mining-contaminated borehole water impairs the entire male reproductive axis: partial mitigation by alpha-lipoic acid in Wistar rats

Festus Tooohukwu Okposhi¹, Collins Nduka Esomchi^{2*}, Kenneth Tochukwu Onwe¹, Leslie Ogechi Onumajuru¹, Ernest Maduabuchi Omeje¹, Favour Chioma Onwumere¹, Precious Chidiogo Ugwu¹, Goodness Emmanuella Onukogu¹, Jennifer Obianuju Olisaeke¹, Chinwendu Princess Victor¹, Chiemerie Promise Omeh¹ and Bianca Adanna Onele¹

¹Department of Anatomy, Faculty of Basic Medical Sciences, Alex Ekwueme Federal University Ndufu Alike, Ikwo, Ebonyi State, Nigeria, ²Department of Human Anatomy, Faculty of Basic Medical Sciences, State University of Medical and Applied Sciences, Igbo Eno, Enugu State, Nigeria

KEYWORDS

alpha-lipoic acid, borehole water, heavy metal contamination, HPG axis, male reproductive toxicity, oxidative stress

A Correction on

Chronic exposure to mining-contaminated borehole water impairs the entire male reproductive axis: partial mitigation by alpha-lipoic acid in Wistar rats

by Okposhi FT, Esomchi CN, Onwe KT, Onumajuru LO, Omeje EM, Onwumere FC, Ugwu PC, Onukogu GE, Olisaeke JO, Victor CP, Omeh CP and Onele BA (2026). *Front. Toxicol.* 8:1988528. doi: 10.3389/ftox.2026.1988528

There was a mistake in the captions of Figures 2, 3, and 4 as published. The annotation keys for the hypothalamic, pituitary, and testicular photomicrographs were omitted. In addition, a production comment was incorporated into the published caption of Figure 4. The figures themselves are unchanged. The corrected captions appear below.

There was a mistake in the caption of Figure 2 as published. The annotation keys were omitted. The corrected caption of **Figure 2** appears below.

“Figure 2 | Representative photomicrographs of the hypothalamus from the experimental groups: (A) Control, (B) Heavy Metal + Recovery, (C) ALA Only, (D) Post-Treatment, and (E) Pre-Treatment. H&E stain; ×400. Scale bar = 50 μm. Arrow, neuronal degeneration; arrowhead, cytoplasmic vacuolation.”

There was a mistake in the caption of Figure 3 as published. The annotation keys were omitted. The corrected caption of **Figure 3** appears below.

“Figure 3 | Representative photomicrographs of the pituitary gland from the experimental groups: (A) Control, (B) Heavy Metal + Recovery, (C) ALA Only, (D) Post-Treatment, and (E) Pre-Treatment. H&E stain; $\times 400$. Scale bar = 50 μm . Hollow arrow, reduced cellular density; green arrowhead, intercellular oedema; dark circle, densely packed endocrine cells; blue arrow, basophils; arrow, cellular degeneration; black arrowhead, vacuolation.”

There was a mistake in the caption of Figure 4 as published. The annotation keys were omitted, and a production comment was added to the caption. The corrected caption of **Figure 4** appears below.

“Figure 4 | Representative photomicrographs of the testes from the experimental groups: (A) Control, (B) Heavy Metal + Recovery, (C) ALA Only, (D) Post-Treatment, and (E) Pre-Treatment. H&E stain; $\times 400$. Scale bar = 50 μm . Black block, degeneration of

seminiferous tubules; hollow star, loss of germinal epithelium; white arrowhead, cytoplasmic vacuolation; yellow arrowhead, interstitial tissue changes.”

The original version of this article has been updated.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.